FAIR DATA ENGINEERING – 2026

FAIRIFICATION OF PREMIER LEAGUE DATA

PLAYER STATISTICS FOR 2024-2025 SEASON

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1. INTRODUCTION

The original dataset, Premier League data from 2016 to 2025 was obtained from Kaggle (https://www.kaggle.com/datasets/danielijezie/premier-league-data-from-2016-to-2024). From this collection, two CSV files were selected: premier_player_info.csv and player_stats_2024_2025_season.csv (https://github.com/giorgosoratis/FAIR-project-group9/raw/main/data/original/original_dataset.zip). The former contains information about players, while the latter contains player performance statistics for the 2024-2025 season.

The original dataset is distributed under the MIT License, which permits the use, modification, and redistribution of the licensed material, provided that the required copyright and permission notices are retained.

The two files were combined into a single dataset by selecting relevant columns from each file and merging them based on player information to create a unified structure. The resulting dataset contains 1,116 player records and 10 columns. The attributes include player names, nationalities, clubs, playing positions, dates of birth, preferred foot, expected assists (xA), expected goals (xG), assists and goals.

The main objective of the project was to improve the dataset's compliance with the FAIR principles: Findability, Accessibility, Interoperability and Reusability. By integrating player information and performance statistics and associating selected attributes with external vocabularies, the project aimed to establish a more semantically meaningful dataset that could support data discovery, integration and reuse.

2. FAIRIFICATION PLANNING

The FAIRification planning phase establishes the objectives, target resources, stakeholders, requirements, semantic types, and competency questions that will guide the FAIRification of the Premier League dataset.

2.1 FAIRIFICATION GOALS

The main goal of this project is to improve the FAIRness of the Premier League player dataset by integrating player information and performance statistics, enriching the data with semantic identifiers, converting it into a machine-readable RDF representation, and publishing the resulting resources with appropriate metadata.

The FAIRification goals are as follows:

1. **Integrate player information and performance statistics:** The retrieved raw data should be combined into a single dataset csv file and then be processed to shape the final dataset to be FAIRified.
2. **Improve the semantic identification of entities:** Reduce ambiguity in selected attribute values by linking player positions, clubs, nationalities to corresponding entities, which enables integration with external datasets.
3. **Represent the dataset in a machine-readable semantic format:** The prepared dataset should be transformed into RDF using an appropriate data model. This will make the relationships between players and their attributes explicit.
4. **Improve the findability of the dataset and its semantic resources:** Descriptive metadata should be provided for both the FAIRified and raw data and persistent and resolvable identifiers should be used for the FAIRified data and the created ontologies.
5. **Improve access to the FAIRified resource:** Both the FAIRified and original dataset, as well as their metadata, should be published in an online repository (github) and provide access locations that allow users to retrieve the resources.
6. **Describe the dataset using structured semantic metadata:** Create metadata that document the dataset's details using a suitable model, that will contribute to the understanding of what the resource contained and how it can be accessed.

Together, these goals define the intended outcomes of the project. Their feasibility and implementation will depend on the available tools, project time, and the technical requirements of the selected dataset. The final FAIRness assessment will determine the extent to which the goals have been achieved.

2.2 TARGET RESOURCE(S)

The target resource is the dataset resulting from the merge of two source files.

The two source files are:

1. premier_player_info.csv, containing descriptive information about football players.
2. player_stats_2024_2025_season.csv, containing player performance statistics for the 2024-2025 season.

The files are combined into a unified dataset, that contains 1,116 records. The resulting dataset was then modified to align with the project's guidelines, retaining only the required attributes for further analysis.

2.3 TARGET STAKEHOLDERS

The identified stakeholders of the project are the following:

1. Student Project Team (Group 9): Responsible for dataset selection, data wrangling and preparation, data reconciliation, conversion to RDF, and the creation and publication of metadata.

2.4 REUSE STAKEHOLDERS

The potential users who may benefit from the FAIRified dataset are the following:

1. **Football analysts and sports statisticians** who can use the dataset to compare player statistics for 2024-2025 season
2. **Football researchers and university students** who can use the dataset for research and educational activities
3. **Football clubs and scouting analysts** who may use the dataset for exploratory player comparison and statistical analysis
4. **Data scientists and developers** who can integrate the dataset into analytical workflows or combine it with other football related datasets and knowledge graphs

These stakeholders may benefit from the FAIRified dataset, since the link to external datasets expands the knowledge field they can acquire compared to the raw resource

2.5 FAIRIFICATION REQUIREMENTS

The FAIRification process heavily depends on several requirements and constraints that influence the choice of tools, the scope of the project and the implementation of FAIR principles.

Deadline	The project should be completed before the given deadline and therefore we focus on manageable dataset consisting of the previously mentioned csv files.
Data access and licensing	The original dataset must be accessible for processing and reuse. Its source and applicable license must be documented to ensure that conditions for redistribution are understood.
Available tools and technical expertise	• OpenRefine for attribute reconciliation with other entities • OpenRefine RDF extension for RDF conversion • Protégé for ontology creation • Contour for metadata schema creation • FDP for metadata creation
Interoperability requirements	The FAIRified dataset should use RDF to represent the data in a machine-readable form. Selected attributes should be linked to appropriate entities through URIs. Where possible, established vocabularies and semantic terms should be reused to facilitate integration with other linked-data resources.
Identification and publication	The FAIRified and original dataset together with their metadata should be published on a GitHub repository, and their redirections must be configured. Any project-related ontology created, must have appropriate W3ID identifiers.

Table 1: FAIRification Requirements

2.6 SEMANTIC TYPES

In the table below, the main concepts represented in the dataset are described. These semantic types establish the domain scope and help identify the entities, attributes and relationships that need to be represented in the semantic data model

Semantic Type	Description
Football Player	A person represented in the dataset as a football player
Football Club	A sports organization associated with a player in the dataset
Playing Position	A football position assigned to a player
Nationality	The recorded nationality of a player
Preferred Foot	The foot recorded as preferred by a player
Date of Birth	A temporal value representing a player's date of birth
Expected Goals	A performance measure representing the expected goals value recorded for a player
Expected Assists	A performance measure representing the expected assists value recorded for a player
Goals	A performance statistic representing the number of goals recorded for a player.
Assists	A performance statistic representing the number of assists recorded for a player.

Table 2: Identified Semantic Types

2.7 COMPETENCY QUESTIONS

Competency questions are the questions that the FAIRified dataset should be able to answer. Below there are some possible questions listed

CQ1. Which players were playing in the Premier League for the 2024-2025 season?
CQ2. Which club did a specific player played for during the 2024-2025 season?
CQ3. What is the playing position of a specific player?
CQ4. What is the nationality of a specific player?
CQ5. What is the date of birth of a specific player?
CQ6. Which players have a particular preferred foot?
CQ7. Which players share the same nationality?
CQ8. How many goals did a specific player score during the 2024-2025 season?
CQ9. Which player scored the most goals?
CQ10. Which player(s) scored no goals during the season, if any?
CQ11. Which player recorded the most assists?
CQ12. Which player recorded the highest expected goals / assists?
CQ13. How many assists did a specific player record?
CQ14. What are the recorded XG and XA values for a specific player?
CQ15. Which players have a higher number of goals than a specified player?
CQ16. Which players have more expected goals / assists than actual goals / assists?

CQ17. Which club has the highest combined number of goals / assists among its players?
CQ18. Which club has the highest combined expected goals / assists among its players?
CQ19. Which players have the same recorded playing position and club?

3. FAIRIFICATION PROCESS

3.1 INITIAL FAIRNESS ASSESSMENT

Principle #	Criteria	Conformance	Note
F1	Globally unique identifier	yes	The Kaggle dataset has a specific URL that identifies its dataset page
F1	Persistent identifier	no	No explicitly verified persistent identifier has been established
F2	Has metadata	yes	Provides basic descriptive metadata, including the dataset title, uploader and description
F2	Has rich metadata	partially	Some descriptive metadata is available, but detailed information about the structure and meaning of the data is limited.
F3	Has data identifier in metadata	no	The metadata does not explicitly include a data identifier for the dataset
F3	Has data identifier with clear predicate in metadata	no	Does not explicitly associate a data identifier with the dataset
F4	Indexed/registered in search resource	yes	Published on Kaggle
A1	Has resolution protocol	yes	Is accessible through a web URL using https
A1.1	Protocol is open, free and universally implementable	yes	HTTPS is an open, widely implemented protocol that can be used without proprietary software
A1.2	Protocol has authentication and authorisation mechanisms	yes	Kaggle supports authentication and access controls
A2	Metadata has longer persistency than data	no	Kaggle maintains a separate dataset page containing descriptive metadata, but there is no confirmed guarantee that this metadata will remain available longer than the dataset files

I1	Appropriate language for knowledge representation	no	The dataset is provided in CSV format and does not explicitly use a formal knowledge-representation language such as RDF.
I2	Vocabularies that follow FAIR principles	no	The original files do not demonstrate the use of shared, FAIR-aligned semantic vocabularies for their fields and values
I3	(meta)data include qualified references to other meta(data)	no	No referneces to external vocabularies or other (meta)data exists
R1	(meta)data are richly described with a plurality of accurate and relevant attributes	partially	The dataset contains relevant attributes but no detailed definitions and documentation is provided
R1.1	Clear and accessible data usage licence	yes	The Kaggle metadata explicitly identifies the MIT licence and provides a link to the license text
R1.2	Detailed provenance	yes	Explicit report of the source and the collection methodology is provided
R1.3	Domain relevant community standards	no	The dataset does not follow formal domain-specific community standards for data structure, terminology or representation

Table 3: Initial FAIRness assessment

3.2 DATA PREPARATION

During the data cleaning and wrangling stage, the two original CSV files were combined into a single dataset. The columns were reviewed and some of them were removed, because including them would make project unnecessarily complex without adding significant value to its specific goals.

The resulting dataset was then prepared for FAIRification using OpenRefine. Through the reconciliation process, relevant data values were matched to corresponding Wikidata entities, and the associated Wikidata URIs were added to the dataset to improve semantic interoperability. In addition, data types and formats were adjusted where necessary, such as standardising the date of birth values into a consistent date format. These steps helped improve the consistency and machine-readability of the dataset and prepared it for conversion to RDF.

3.3 SEMANTIC DATA MODEL

The semantic data model was designed to represent football player information and performance statistics in a structured, machine-readable form.

Each row in the dataset represents a football player record containing the player's identifying information and performance statistics. Therefore, Football Player is the central entity in the semantic data model, as illustrated in the diagram below. It is connected to related concepts, such as sports clubs, countries, football positions and preferred foot. These external entities are linked to relevant Wikidata concepts enabling the dataset to be connected to external information sources. The model also includes player attributes, including name and date of birth, as well as performance statistics such as goals, assists, expected goals and expected assists.

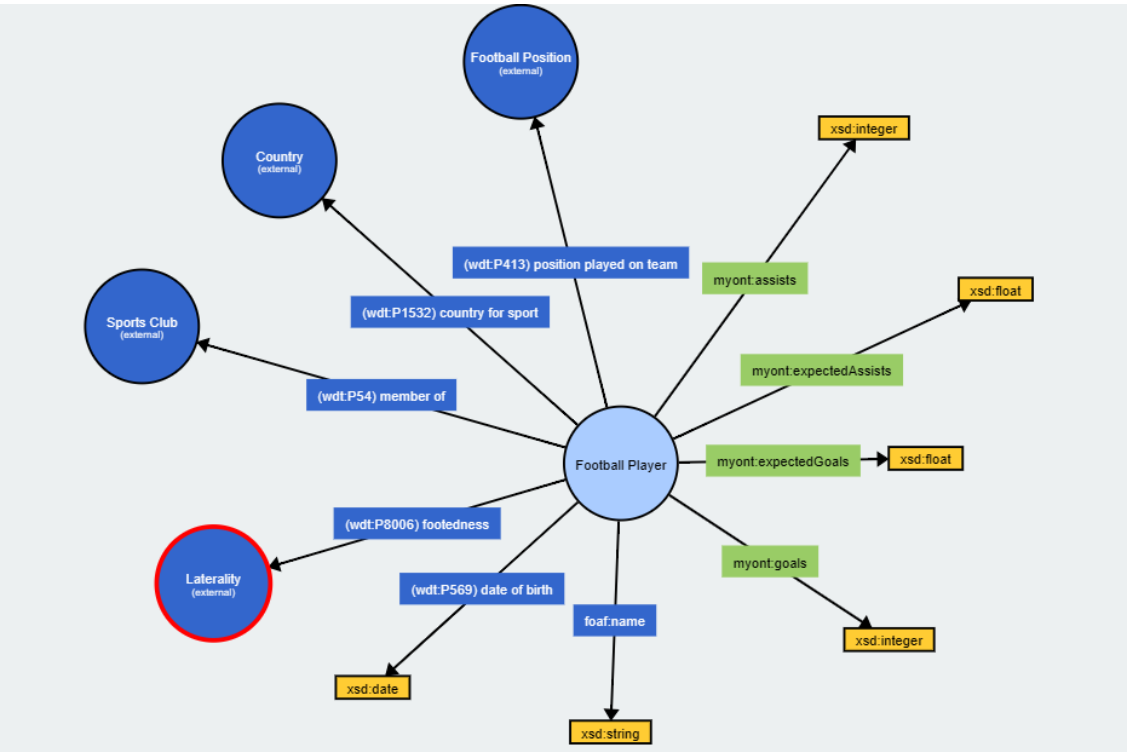


Figure 1: Semantic Data Model Diagram

Entities	URI
Country	https://www.wikidata.org/wiki/Q6256
Football Position	https://www.wikidata.org/wiki/Q4611891
Sports Club	https://www.wikidata.org/wiki/Q847017
Laterality	https://www.wikidata.org/wiki/Q10927834
Properties	**URI**
name	http://xmlns.com/foaf/0.1/name
Position played on a team	https://www.wikidata.org/wiki/Property:P413
Member of	https://www.wikidata.org/wiki/Property:P54
Country for sport	https://www.wikidata.org/wiki/Property:P1532
Footedness	https://www.wikidata.org/wiki/Property:P8006
Date of Birth	https://www.wikidata.org/wiki/Property:P569

Table 4: URIs from existing vocabularies

In addition to reusing concepts and properties from existing vocabularies, the remaining entities and attributes (myont:FootballPlayer, myont:expectedAssists, myont:expectedGoals, myont:assists, myont:goals) required for the dataset were defined in a custom ontology using Protégé. The ontology uses the namespace URI https://w3id.org/FAIR-course-UT/2026/group09/ont#, following the W3ID identifier pattern specified in the course requirements. This provides the structure needed to represent the football player information and performance statistics that were not covered by the selected existing vocabularies. By combining existing vocabulary terms with custom-defined ontology elements, the semantic data model supports the representation of the dataset according to the project's requirements.

3.4 DATA TRIPLIFICATION

After defining the semantic data model and preparing the dataset, the data were transformed from CSV into RDF using the RDF extension in OpenRefine. During the triplification process, each player

record was represented as RDF statements consisting of a subject, predicate and object. An example RDF statement for a single player is shown below.

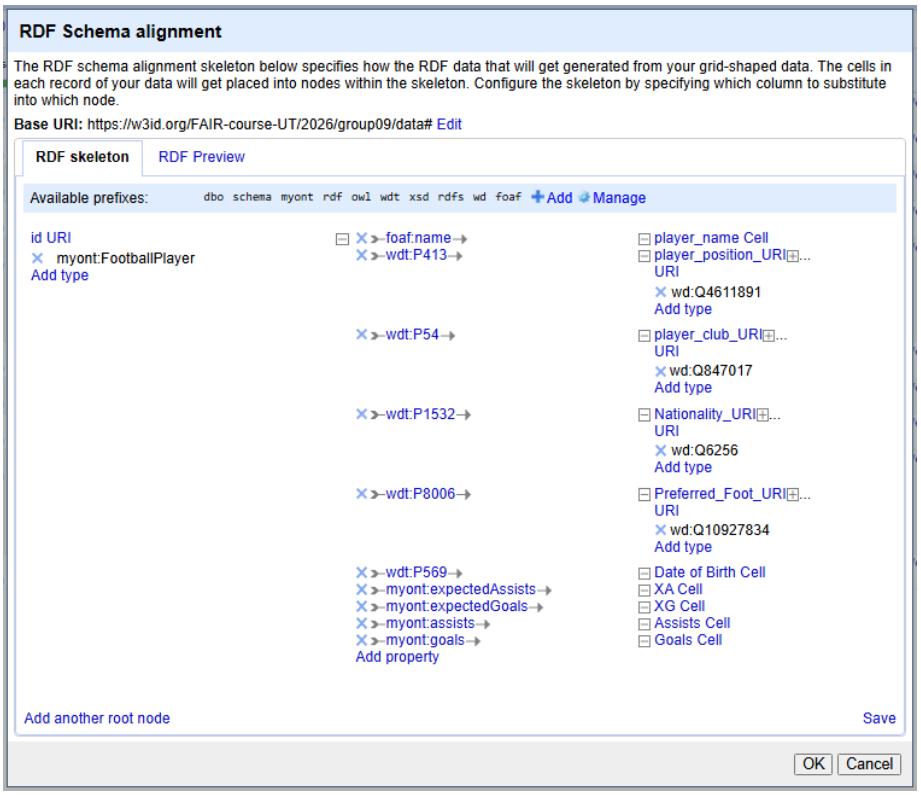


Figure 2: RDF Schema Alignment in OpenRefine.

```
<https://w3id.org/FAIR-course-UT/2026/group09/data#player-9> a myont:FootballPlayer;
foaf:name "Simon Adingra" .
<https://www.wikidata.org/entity/Q280658> a wd:Q4611891 .
<https://w3id.org/FAIR-course-UT/2026/group09/data#player-9> wdt:P413 <https://www.wikidata.org/entity/Q280658> .
<https://www.wikidata.org/entity/Q19453> a wd:Q847017 .
<https://w3id.org/FAIR-course-UT/2026/group09/data#player-9> wdt:P54 <https://www.wikidata.org/entity/Q19453> .
<https://www.wikidata.org/entity/Q1008> a wd:Q6256 .
<https://w3id.org/FAIR-course-UT/2026/group09/data#player-9> wdt:P1532 <https://www.wikidata.org/entity/Q1008> .
<https://www.wikidata.org/entity/Q1843233> a wd:Q10927834 .
<https://w3id.org/FAIR-course-UT/2026/group09/data#player-9> wdt:P8006 <https://www.wikidata.org/entity/Q1843233>;
wdt:P569 "2002-01-01"^^xsd:date;
myont:expectedAssists "2.79"^^xsd:float;
myont:expectedGoals "2.46"^^xsd:float;
myont:assists "2"^^xsd:integer;
myont:goals "2"^^xsd:integer .
```

Figure 3: Example of RDF triples generated through OpenRefine

As can be seen each player is assigned a URI within the project's W3ID namespace and is represented as an instance of the custom myont:FootballPlayer class. The player's name is represented using the foaf:name property, while relationships with external entities, such as playing position, club and country, are represented using Wikidata entities and properties. Performance statistics, including expected assists, expected goals, assists and goals, are represented using the custom properties defined in the project's ontology.

3.5 SEMANTIC METADATA MODEL

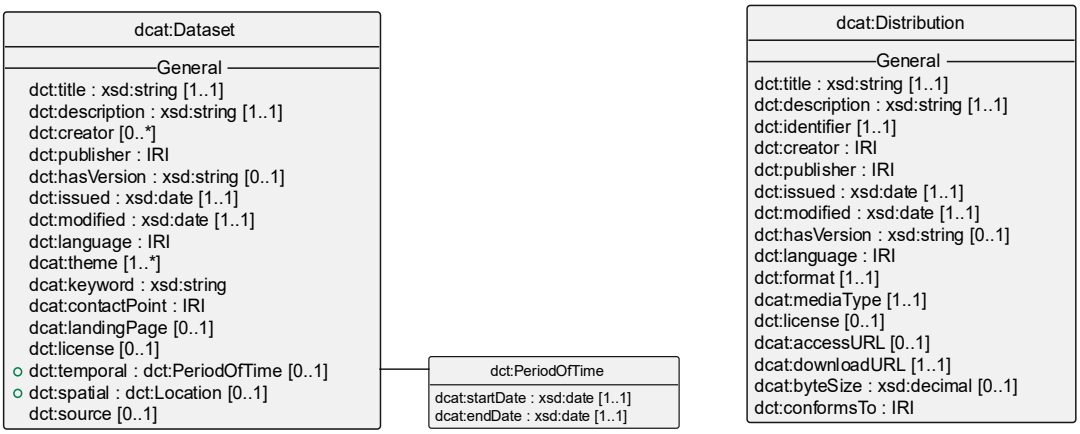


Figure 4: Dataset and Distribution Metadata Schema

The semantic metadata model was defined using SHACL (Shapes Constraint Language) to describe the metadata of both the dataset and its distribution. Two separate SHACL schemas were created: one targeting dcat:Dataset and another targeting dcat:Distribution. This separation allows the metadata describing the dataset as a whole, to be distinguished from the metadata describing its specific data representation and access options.

The dataset schema defines properties such as title, description, creator, publisher, version, publication and modification dates, language, theme, keywords, contact point, landing page, license, temporal and spatial coverage, and source.

The distribution schema describes the specific representation of the dataset and its availability. It includes properties such as title, description, identifier, creator, publisher, publication and modification dates, version, language, format, media type, license, access URL, download URL, byte size and conformance to a standard.

Vocabulary	Namespace URI	Terms used
DCAT	http://www.w3.org/ns/dcat#	dcat:Dataset, dcat:Distribution, dcat:theme, dcat:keyword, dcat:contactPoint, dcat:landingPage, dcat:mediaType, dcat:accessURL, dcat:downloadURL, dcat:byteSize, dcat:startDate, dcat:endDate
DCTERMS	http://purl.org/dc/terms/	dct:title, dct:description, dct:creator, dct:publisher, dct:hasVersion, dct:issued, dct:modified, dct:language, dct:license, dct:temporal, dct:spatial, dct:source, dct:format, dct:identifier, dct:conformsTo
RDF	http://www.w3.org/1999/02/22-rdf-syntax-ns#	rdf:type

XSD	http://www.w3.org/2001/XMLSchema#	xsd:string, xsd:date, xsd:decimal
SHACL	http://www.w3.org/ns/shacl#	sh:NodeShape, sh:PropertyGroup, sh:targetClass, sh:property, sh:path, sh:name, sh:description, sh:nodeKind, sh:datatype, sh:minCount, sh:maxCount, sh:order, sh:group, sh:class, sh:node
DASH	https://datashapes.org/dash	dash:editor, dash:TextFieldEditor, dash:TextAreaEditor, dash:URIEditor, dash:DatePickerEditor, dash:DetailsEditor, dash:LiteralViewer, dash:LabelViewer

Table 5: Vocabularies, namespace URIs and terms used in semantic metadata model

The SHACL code for both the metadata and distribution schema can be found here: https://github.com/giorgosoratis/FAIR-project-group9/tree/main/metadata/schemas

3.6 METADATA AND DATA PUBLICATION

After the data FAIRification process metadata and data were published on a GitHub repository.

Links to the published resources:

1. **Original Data:** https://github.com/giorgosoratis/FAIR-project-group9/tree/main/data/original
2. **FAIRified RDF Data:** https://github.com/giorgosoratis/FAIR-project-group9/tree/main/data/fairified
3. **Metadata Records:** https://github.com/giorgosoratis/FAIR-project-group9/tree/main/metadata/records
4. **Ontology:** https://github.com/giorgosoratis/FAIR-project-group9/tree/main/ontology

Persistent identifiers (W3ID):

- Original Data: https://w3id.org/FAIR-course-UT/2026/group09/data-original
- FAIRified RDF Data: https://w3id.org/FAIR-course-UT/2026/group09/data
- Ontology: https://w3id.org/FAIR-course-UT/2026/group09/ont

3.7 FINAL FAIRNESS ASSESSMENT

Principle #	Criteria	Conformance	Note
F1	Globally unique identifier	yes	W3ID is used, which provides globally uniqu, persistent and resolvable identifiers
F1	Persistent identifier	yes	
F2	Has metadata	yes	Metadata is provided to describe the dataset and its main characteristics
F2	Has rich metadata	yes	The metadata includes descriptive information about the dataset, such as its title, description, creator, source, licence, and temporal and spatial coverage

F3	Has data identifier in metadata	yes	The metadata includes an identifier or reference associated with the data resource
F3	Has data identifier with clear predicate in metadata	yes	The metadata is intended to identify the data resource explicitly. Confirm that the identifier is present and correctly associated in the published records.
F4	Indexed/registered in search resource	partially	Published on GitHub; indexing in searchable resources is unverified.
A1	Has resolution protocol	yes	The resource can be accessed through a web-based protocol using its URI
A1.1	Protocol is open, free and universally implementable	yes	HTTP/HTTPS provides an open and widely implemented protocol for accessing the resource
A1.2	Protocol has authentication and authorisation mechanisms	yes	GitHub supports authentication and authorization for restricted resources.
A2	Metadata has longer persistency than data	partially	Metadata are stored separately on GitHub and published on a local FDP. Public deployment of the FDP would support metadata persistence
I1	Appropriate language for knowledge representation	yes	RDF is used to represent the data in a structured, machine-readable format
I2	Vocabularies that follow FAIR principles	yes	Established vocabularies, including DCAT and Dublin Core Terms, are used to describe the dataset and its distribution
I3	(meta)data include qualified references to other meta(data)	yes	Selected dataset attributes are linked to Wikidata entities through URIs, providing references to external structured knowledge
R1	(meta)data are richly described with a plurality of accurate and relevant attributes	yes	The metadata contains relevant descriptive attributes, including the dataset's title, description, creator, source, licence, and coverage
R1.1	Clear and accessible data usage licence	yes	License is explicitly specified in the metadata records.
R1.2	Detailed provenance	yes	Source, data preparation, reconciliation and RDF transformation are documented.
R1.3	Domain relevant community standards	partially	The ontology uses football-specific concepts, but the report does not establish that a recognised community standard is followed.

Table 6: Final FAIRness assessment

4. DISCUSSIONS AND CONCLUSIONS

This project focused on applying FAIRification principles to Premier League player statistics for the 2024/2025 season. The original data were cleaned and transformed into RDF/Turtle, enabling a structured, machine-readable representation. Selected entities, including player positions, clubs, nationalities and preferred-foot categories, were reconciled with Wikidata using OpenRefine, establishing links to external knowledge resources.

A project-specific ontology was developed to represent football players and their statistical attributes. In addition, structured metadata was created for the catalog, dataset and distributions, supported by SHACL schemas defining the expected metadata structure and constraints.

The main challenges involved entity reconciliation and semantic modelling, particularly selecting appropriate existing vocabularies and determining which properties required project-specific definitions. These challenges were addressed by reusing existing vocabularies where appropriate and defining custom properties for player statistics. The project demonstrated that FAIRification requires not only data transformation but also careful preparation, semantic modelling and metadata definition.

Overall, the project improved the dataset's potential for findability, interoperability and reuse through persistent identifiers, external entity links and structured metadata. Future work could focus on linking the dataset with additional football-related datasets, extending its coverage to more seasons and incorporating additional player statistics to support broader comparisons and analysis.

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